# Carlos Mauricio Peredo, Ph.D.

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## Professional Summary

Ph.D.-trained quantitative researcher turned AI product leader, architecting and shipping production AI systems, with a specialty in regulated industries. I generate over $350,000 annually in signed AI engagements, spanning live production voice AI agents and the governance platform that now powers AI delivery firmwide.

## Work Experience

**Lead AI & Data Architect** | 2025–current

Vivitec AI, Vivitec Inc.

- Architected & deployed AI solutions for small-to-medium businesses, including data lake infrastructure, agentic products, & SaaS applications that automate workflows & surface actionable insights
- Delivered a production retrieval-augmented generation (RAG) platform for a signed $135,500 engagement, indexing client bid documents with quote-level provenance for AI-assisted analysis
- Led enterprise AI adoption initiatives, facilitating executive roundtables, developing governance frameworks, & advising on strategic AI roadmaps
- Generated over $350,000 in consulting revenue while owning delivery end-to-end

**Senior Research Scientist** | 2024–2026

Student Success Analytics, Western Governors University

- Developed & deployed predictive models to optimize financial aid, resulting in 12% reduction in student attrition
- Led design & rollout of a "Last Mile" Financial Aid pilot, projected to mitigate 42% of at-risk completions
- Implemented an LLM-based tool to analyze root causes of student attrition, enabling targeted interventions

**Program Manager & Head of Research** | 2022–2024

Peredo Comparative Morphology Lab, Miami University

- Directed a 6–10 person research team producing peer-reviewed technical publications
- Engineered a hybrid Neural Network–Random Forest ensemble to analyze fragmentary, high-missingness data
- Built a Computer Vision pipeline (image classification, object detection) for research inventory automation

**Program Manager & Senior Researcher** | 2019–2022

Peredo Comparative Morphology Lab, University of Michigan

- Founded and directed a 3–5 person research lab as sole PI, producing peer-reviewed technical publications
- Designed & standardized data collection workflows for research reproducibility
- Secured & led NSF and university fellowship funding as sole PI, managing personnel, timelines, & deliverables

**Research Scientist** | 2014–2019

Smithsonian NMNH & George Mason University

- Directed an international science diplomacy initiative with Qatar's Ministry of Culture and served as scientific consultant for PBS/BBC educational programming
- Standardized protocols for collecting, processing, & archiving 100+TB of 3D data across 1M+ row datasets

## Education

- **Ph.D.,** Quantitative Evolutionary Ecology, George Mason University
- **M.S.,** Quantitative Evolutionary Ecology, George Mason University
- **B.S.,** Applied Biology, Seton Hill University
- **Certificate,** Business Leadership, Western Governors University
- **Certificate,** Project Management, Western Governors University

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## Skills

### Data Science & Programming

- **Languages & Tools:** R, Python, SQL, React, TypeScript, Databricks, GitHub, Jupyter, Docker, Claude Code
- **AI/ML Techniques:** LLMs, Agentic & Multi-Agent Systems, MCP Server Development, RAG, Voice AI & Conversational Agents, Neural Networks, Random Forests, Computer Vision
- **Cloud & Deployment:** Microsoft Azure (Functions, Container Apps, AI Search, Bicep IaC), RBAC & Access Control, Google Cloud Platform (GCP)

### Leadership & Strategy

- **People:** Leadership, Coaching & Mentoring, Cross-Functional Collaboration, Building & Supervising Teams
- **Strategy:** Roadmapping, Strategic Planning, Organizational Alignment, AI Governance, Change Management
- **Communication:** Stakeholder Engagement, Executive Communication, Recommending Action
- **Business Development:** Client Relations, Revenue Generation, Consulting

### Statistics & Research Methods

- **Modeling & Inference:** Predictive Modeling, Causal Inference, Bayesian Inference
- **Research Design:** Experimental Design, A/B Testing, Mixed Methods Research, Survey Design

### Visualization & Design

- **Data Visualization:** Tableau, Power BI, LucidChart, Custom Dashboarding
- **Design & Modeling:** Adobe Creative Suite, Autodesk Maya, Artec Studio, Mimics, Fusion360

## Industry Certifications

- **Data & AI:** Databricks Fundamentals, Databricks Lakehouse Fundamentals, Databricks Generative AI Fundamentals, Microsoft Data Fundamentals
- **AI Platforms:** Anthropic Certified (Agentic Systems, MCP Development, Claude Code, API Integration)
- **Research & Leadership:** Ethical Management of Human-Sourced Datasets, Working with IACUC

## Languages

- **English:** native; fluent in academic and technical communication
- **Spanish:** native; fluent in academic and technical communication

## Research & Funding

- 28 peer reviewed research articles (12 first authored)
- 26 conference abstracts (16 first authored)
- Over $500,000 USD awarded in external research funding
- Over $350,000 USD generated through consulting engagements
- Created estimated $750,000+ in efficiency gains through automation & process optimization

## Leadership Philosophy

**I believe in evidence over instinct.**
Let's think big and take calculated risks. Data and AI help us get it right, not just get there fast.

**I believe in creating clarity from chaos.**
Let's cut through ambiguity, tell a compelling story, and chart the path forward.

**I believe AI systems carry real consequences.**
Behind every dataset and every model are real people. Let's build and govern it with care, ethics, and integrity.

**I believe in putting people first.**
Curiosity fuels ingenuity. Let's build teams that stay curious and lead with compassion.